



**CALICUT UNIVERSITY – FOUR-YEAR UNDER  
GRADUATE PROGRAMME (CU-FYUGP)**

**BSc PHYSICS HONOURS**

|                   |                                                                                                                                                                                                    |                     |                      |                       |                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------|-----------------------|----------------|
| Programme         | <b>B.Sc. Physics Honours</b>                                                                                                                                                                       |                     |                      |                       |                |
| Course Title      | <b>ELECTRONICS II</b>                                                                                                                                                                              |                     |                      |                       |                |
| Type of Course    | <b>Core in Major</b>                                                                                                                                                                               |                     |                      |                       |                |
| Semester          | <b>VI</b>                                                                                                                                                                                          |                     |                      |                       |                |
| Academic Level    | <b>300 - 399</b>                                                                                                                                                                                   |                     |                      |                       |                |
| Course Details    | Credit                                                                                                                                                                                             | Lecture<br>per week | Tutorial<br>per week | Practical<br>per week | Total<br>Hours |
|                   | 4                                                                                                                                                                                                  | 3                   | -                    | 2                     | 75             |
| Pre-requisites    | PHY2CJ101- Electronics I                                                                                                                                                                           |                     |                      |                       |                |
| Course<br>Summary | Course provides students with a comprehensive understanding of transistor operation, FET characteristics, and Op-Amp applications, preparing them for designing and analyzing electronic circuits. |                     |                      |                       |                |

**Course Outcomes (CO):**

| CO  | CO Statement                                                             | Cognitive<br>Level* | Knowledge<br>Category# | Evaluation Tools<br>used |
|-----|--------------------------------------------------------------------------|---------------------|------------------------|--------------------------|
| CO1 | Understand the fundamental principles of analog and digital electronics. | Understand          | Basic Concepts         | Quizzes, Tests           |
| CO2 | Analyse different types of amplifiers and their applications.            | Analyse             | Applications           | Homework<br>Assignments  |

|                                                                                                                                                                                                   |                                                                                   |            |                  |                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------|------------------|------------------------|
| CO3                                                                                                                                                                                               | Design amplifier circuits based on given specifications.                          | Apply      | Circuit Design   | Laboratory Experiments |
| CO4                                                                                                                                                                                               | Analyse the operation of different types of FETs (JFETs, MOSFETs).                | Analyse    | Device Operation | Homework Assignments   |
| CO5                                                                                                                                                                                               | Understand the operational principles of Operational Amplifiers (Op-amps).        | Understand | Basic Concepts   | Quizzes, Assignments   |
| CO6                                                                                                                                                                                               | Analyse and design sequential logic circuits using state diagrams and flip-flops. | Analyse    | Circuit Design   | Laboratory Experiments |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)<br>Metacognitive Knowledge (M) |                                                                                   |            |                  |                        |

**Detailed Syllabus:**

| Module    | Unit                                                                                 | Content                                                  | Hrs<br>(45<br>+30) | Marks<br>(70) |
|-----------|--------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------|---------------|
| <b>I</b>  | <b>TRANSISTOR AMPLIFIERS AND OSCILLATORS</b>                                         |                                                          | <b>12</b>          | <b>20</b>     |
|           | 1                                                                                    | Single Stage Transistor Amplifier                        | 3                  |               |
|           | 2                                                                                    | Multi stage amplifier                                    | 3                  |               |
|           | 3                                                                                    | Feed Back                                                | 2                  |               |
|           | 4                                                                                    | Advantages of negative feedback                          | 1                  |               |
|           | 5                                                                                    | Transistor Oscillators – Colpitt's, Hartley, Phase shift | 3                  |               |
|           | Sections 10.1 – 10.6, 11.3 – 11.5, 13.1 – 13.6, 14.3 – 14.7, 14.10 – 14.13 of Book 1 |                                                          |                    |               |
| <b>II</b> | <b>FIELD EFFECT TRANSISTORS</b>                                                      |                                                          | <b>10</b>          | <b>15</b>     |
|           | 6                                                                                    | Types of FET                                             | 1                  |               |
|           | 7                                                                                    | Principle and working of JFET                            | 2                  |               |

|            |                                                                                        |                                                 |           |           |
|------------|----------------------------------------------------------------------------------------|-------------------------------------------------|-----------|-----------|
|            | 8                                                                                      | Difference Between JFET and BJT                 | 1         |           |
|            | 9                                                                                      | JFET amplifier                                  | 2         |           |
|            | 10                                                                                     | Output Characteristics of JFET                  | 2         |           |
|            | 11                                                                                     | MOSFET                                          | 2         |           |
|            | Sections 19.1 – 19.12, 19.27 – 19.28 of Book 1                                         |                                                 |           |           |
| <b>III</b> | <b>OPERATIONAL AMPLIFIERS</b>                                                          |                                                 | <b>12</b> | <b>20</b> |
|            | 12                                                                                     | Differential Amplifier                          | 3         |           |
|            | 13                                                                                     | OP-Amp                                          | 2         |           |
|            | 14                                                                                     | OP-Amp with Negative Feedback                   | 3         |           |
|            | 15                                                                                     | Summing Amplifiers                              | 2         |           |
|            | 16                                                                                     | Integrator and Differentiator                   | 2         |           |
|            | Sections 25.1 – 25.5, 25.7, 25.8, 25.15 – 25.17, 25.22 – 25.27, 25.32 – 25.37 of Book1 |                                                 |           |           |
| <b>IV</b>  | <b>DIGITAL ELECTRONICS</b>                                                             |                                                 | <b>11</b> | <b>15</b> |
|            | 17                                                                                     | Basic Logic Gates (26.11-26.14)                 | 2         |           |
|            | 18                                                                                     | Combination of Logic Gates (26.15-26.17, 26.19) | 1         |           |
|            | 19                                                                                     | Boolean Algebra and logic circuits(26.20-26.23) | 2         |           |
|            | 20                                                                                     | Combinational logic circuits(26.24-26.30)       | 2         |           |
|            | 21                                                                                     | Electronic Adders (26.32 )                      | 1         |           |
|            | 22                                                                                     | Flip – Flops (26.33)                            | 3         |           |
|            | Sections 26.11 – 26.17, 26.19 – 26.30, 26.32, 26.33 of Book1                           |                                                 |           |           |
| <b>V</b>   | <b>PRACTICALS</b>                                                                      |                                                 | <b>30</b> |           |

|   |                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|   | Conduct any 6 experiments from the given list (4 experiments from 1-8 and 2 from 9-12) and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 <sup>th</sup> experiment may also be selected from the given list.                                                                                                                                          |  |  |
| 1 | <b>Study the frequency response of common emitter(CE) transistor amplifier.</b> <ul style="list-style-type: none"> <li>Design a CE transistor amplifier of a given gain (mid-gain) using voltage divider bias.</li> <li>Analyse the frequency response, draw the curve and find the bandwidth, without feedback.</li> </ul>                                                                                   |  |  |
| 2 | <b>Study the negative feedback in CE transistor amplifier.</b> <ul style="list-style-type: none"> <li>Design a CE transistor amplifier of a given gain (mid-gain) using voltage divider bias.</li> <li>Determine the voltage gain with and without negative feedback.</li> <li>Repeat the analysis by changing the feedback fraction.</li> <li>Optional: Frequency response study may be repeated.</li> </ul> |  |  |
| 3 | <b>Construction of LC oscillator (Hartley or Colpitt's)</b> <ul style="list-style-type: none"> <li>Construct a LC oscillator (Hartley or Colpitt's) and measure the frequency using CRO/ExpEYES for different values of L and C. Compare with the theoretical values.</li> </ul>                                                                                                                              |  |  |
| 4 | <b>Construction of phase shift oscillator</b> <ul style="list-style-type: none"> <li>Construct a phase shift oscillator and measure the frequency using CRO/ExpEYES for different values of R and C. Compare with the theoretical values.</li> </ul>                                                                                                                                                          |  |  |
| 5 | <b>Construction astable multivibrator using transistors.</b> <ul style="list-style-type: none"> <li>Construct an astable multivibrator using transistors and measure the frequency using CRO/ExpEYES for different values of R and C. Compare with the theoretical values.</li> </ul>                                                                                                                         |  |  |
| 6 | <b>Construction astable multivibrator using IC 555.</b> <ul style="list-style-type: none"> <li>Design an astable multivibrator of desired frequency (say 1000 Hz) and duty cycle (say 60%) using IC 555 and measure the frequency using CRO/ExpEYES. Compare with the theoretical values.</li> </ul>                                                                                                          |  |  |
| 7 | <b>Operational Amplifier –inverting, non inverting amplifier and voltage follower.</b> <ul style="list-style-type: none"> <li>Design inverting and non inverting amplifiers of different voltage gain.</li> <li>Measure and verify the gain using CRO/ExpEYES.</li> <li>Construct a voltage follower and verify that the gain is unity.</li> </ul>                                                            |  |  |

|    |                                                                                                                                                                                                                                                                                                                                     |  |  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| 8  | <b>Operational Amplifier- adder, subtractor</b> <ul style="list-style-type: none"> <li>Design arithmetic circuits(adder and subtractor) using OP AMP, with two input voltages and measure the result using multimeter/CRO/ExpEYES.</li> </ul>                                                                                       |  |  |
| 9  | <b>Digital electronics</b><br><b>Construction of basic gates using diodes (AND, OR) &amp; transistor (NOT)</b> <ul style="list-style-type: none"> <li>Realize the logic AND and OR gates using diodes and NOT gate using a transistor and verify the truth table. Logic output can be checked using a multimeter or LED.</li> </ul> |  |  |
| 10 | <b>Construct Half adder using universal gates and study the operation.</b> <ul style="list-style-type: none"> <li>Implement half adder using NAND/NOR gates and verify the truth table for each input/output combination.</li> </ul>                                                                                                |  |  |
| 11 | <b>Verification of De-Morgan's Theorems using basic gates.</b> <ul style="list-style-type: none"> <li>Realize the either side of the De-Morgan's Theorems using gates from appropriate ICs and verify the truth table for each input/output combination.</li> </ul>                                                                 |  |  |
| 12 | <b>To construct and study the operations of the RS and JK Flip-Flops using IC's</b> <ul style="list-style-type: none"> <li>Realize RS Flip-Flop using NAND gates and verify the truth table</li> <li>Realize JK Flip-Flop using NAND gates from appropriate ICs and verify the truth table</li> </ul>                               |  |  |

## Books and References:

1. Principles of Electronics by V K Mehtha and Rohith Mehtha (Book 1)
2. Electronic Devices and Circuit Theory by Robert L. Boylestad and Louis Nashelsky
3. Electronic Principles by Albert Malvino and David J. Bates
4. Analog Electronics: Devices, Circuits, and Techniques by Chitralekha Mahanta
5. Basic Electrical and Electronics Engineering by R.K. Rajput
6. Semiconductor Devices: Physics and Technology by S. M. Sze

**Mapping of COs with PSOs and POs :**

|      | PSO 1 | PSO 2 | PSO 3 | PSO 4 | PSO 5 | PSO 6 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|------|-------|-------|-------|-------|-------|-------|-----|-----|-----|-----|-----|-----|-----|
| CO 1 | 2     | 3     | 0     | 2     | 1     | 0     | 2   | 1   | 2   | 0   | 2   | 2   | 1   |
| CO 2 | 1     | 2     | 3     | 3     | 2     | 2     | 0   | 1   | 0   | 2   | 1   | 2   | 0   |
| CO 3 | 2     | 1     | 2     | 2     | 1     | 1     | 2   | 1   | 2   | 1   | 0   | 1   | 1   |
| CO 4 | 2     | 2     | 0     | 1     | 2     | 0     | 3   | 2   | 1   | 2   | 1   | 0   | 0   |
| CO 5 | 2     | 0     | 2     | 2     | 1     | 2     | 2   | 0   | 1   | 1   | 2   | 1   | 0   |
| CO 6 | 1     | 2     | 1     | 0     | 2     | 1     | 1   | 2   | 0   | 2   | 1   | 2   | 1   |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| 0     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

**Mapping of COs to Assessment Rubrics**

|      | Internal Theory / Practical Exam | Assignment / Viva | Practical Skill Evaluation | End Semester Examinations |
|------|----------------------------------|-------------------|----------------------------|---------------------------|
| CO 1 | ✓                                | ✓                 |                            | ✓                         |
| CO 2 | ✓                                | ✓                 |                            | ✓                         |
| CO 3 | ✓                                | ✓                 |                            | ✓                         |
| CO 4 | ✓                                | ✓                 |                            | ✓                         |
| CO 5 | ✓                                | ✓                 |                            | ✓                         |
| CO 6 |                                  | ✓                 | ✓                          |                           |